Kernels and Dual Bases to Explore: a Plan

Robin Langer and Claude Code

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1 The Program

The primitive object is a kernel W on a set X together with the *bilinear* pairing $\langle f, g\rangle = \int_X fg$. From it:

1. *polynomials* P_n with $\int_X P_m P_n W = h_n \delta_{mn}$. Positivity of W is **not** required — only nonvanishing of the Hankel determinants of the moments $\int x^n W$. Signed and complex kernels are admitted.
2. an analytic *split* $W = w_+ w_-$ and the dual bases $F_n = w_+ \hat{P}_n$, $G_n = w_- \hat{P}_n$, biorthonormal: $\int F_m G_n = \delta_{mn}$.
3. the *matching matrix* $c_{mk} = \int_X \hat{P}_m \hat{P}_k \, w_+ W$, expanding one basis in the polynomials.

The *orthogonal* case is the self-dual specialisation: $w_+ = w_- = \sqrt{W}$, $G_n = \overline{F_n} = \Phi_n$, and the pairing becomes the L^2 inner product. Hermite (Gaussian) is exactly self-dual; MP taught us the split $\Gamma(\frac{1}{4} + ix) \cdot \Gamma(\frac{1}{4} - ix)$ beats the symmetric root $\sqrt{W}$ — the arithmetic lives in the dual pair, not the orthogonal functions. So the plan admits any kernel with a good split, whether or not a positive orthogonal theory exists.

For each new kernel the questions are always: decay rate of c_{0k}, integrality (over which ring), generating function, and zeros of the truncated approximants.

What the first two instances taught

	Hermite	Meixner–Pollaczek ($\lambda = \frac{1}{4}$)
kernel W	e^{-x^2} (entire)	$\Gamma(\frac{1}{4} + ix)\Gamma(\frac{1}{4} - ix)$ (poles at $\pm i\frac{1}{4}$)
decay b_{0k}	geometric $3^{-k/2}$	power law $k^{-\alpha}$, $\alpha \approx \frac{3}{4}$
integrality	$\mathbb{Z}$, directly	none for $\sqrt{W}$; dual pair $\Rightarrow \mathbb{Z}A \oplus \mathbb{Z}B$
generators	1	$A, B = \sqrt{2e}\, K_0(\frac{1}{2})$, $\sqrt{2e}\, K_1(\frac{1}{2})$
GF	$e^{(\text{quadratic})}$	$e^{\zeta/2} K_0(\zeta/2)$, ζ = product of Cayley transforms

Moral: the family name is irrelevant; the analytic class of the kernel decides everything. The plan below is therefore organised by strata of weight analyticity, with one canonical representative each.

2 The Strata

2.1 Entire kernels *done*

$W = e^{-x^2}$ on $\mathbb{R}$, Hermite H_n. Geometric decay, integer matching matrix, closed GF. The base case.

2.2 One pair of Γ's *done ($\lambda = \frac{1}{4}$)*

$W = |\Gamma(\lambda + ix)|^2$ on $\mathbb{R}$, Meixner–Pollaczek $P_n^{(\lambda)}(x; \frac{\pi}{2})$. Bump–Ng: for $\lambda = \frac{1}{4}$ the functions are Mellin transforms of even Hermite functions — the $\Gamma(s/2)$ factor of ξ. Dual pair restores arithmetic; decay stays slow (pole at finite distance).

2a. The other parameter, $\lambda = \frac{3}{4}$. Same machinery, kernel $|\Gamma(\frac{3}{4} + ix)|^2$ = Mellin side of *odd* Hermite functions = gamma factor $\Gamma(\frac{s+1}{2})$ of odd Dirichlet L-functions. It has already knocked twice: the spurious zeros of $F_0^{(N)}$ converge onto the line $\operatorname{Im} x = -\frac{1}{2}$ (the $\lambda = \frac{3}{4}$ line), and the joint weight collapses:

$$|\Gamma(\tfrac{1}{4} + ix)|^2\, |\Gamma(\tfrac{3}{4} + ix)|^2 \;=\; \frac{2\pi^2}{\cosh 2\pi x}.$$

2b. The joint $\frac{1}{4} \otimes \frac{3}{4}$ matrix. Cross matching coefficients $\int \hat{P}_m^{(1/4)} \hat{P}_k^{(3/4)} \frac{dx}{\cosh 2\pi x}$ (and variants): an *elementary* kernel, sech = its own Fourier transform, poles at $\pm\frac{i}{4}$. Every integral is cheap; this is the fastest route to a genuinely new table. Expect: two-family checkerboard, and a chance of closed forms (the sech moment integrals are classical — Euler numbers).

2.3 Two pairs of Γ's: continuous Hahn *next*

$W = |\Gamma(a+ix)\Gamma(b+ix)|^2$ on $\mathbb{R}$, continuous Hahn $p_n(x; a, b)$. These are the Mellin transforms of *Laguerre* functions — the half-line theory pushed through Bump–Ng. Matching integrals now have 3 or 4 whole Γ-pairs: genuine Barnes integrals. The Kummer-U lemma that cracked the MP dual table sits one step *below* Barnes' first lemma; here Barnes' lemmas apply natively. Prediction: exact closed forms return, integrality over a ring generated by hypergeometric constants at fixed argument.

2.4 Four Γ's: Wilson *after Hahn*

$W = \left|\frac{\prod_j \Gamma(a_j+ix)}{\Gamma(2ix)}\right|^2$, Wilson polynomials $W_n(x^2; a, b, c, d)$ — top of the Askey scheme, orthogonality *is* a Barnes-type integral (de Branges, Wilson). Everything below is a degeneration; if a uniform statement about matching matrices exists, it lives here. High risk, high yield; do after continuous Hahn calibrates expectations.

2.5 Half-line: Laguerre *control*

$W = x^\alpha e^{-x}$ on $(0, \infty)$, Laguerre $L_n^{(\alpha)}$. Kernel entire in the bulk but with a branch point at the endpoint 0 (for $\alpha \notin 2\mathbb{Z}$). Interpolates the Hermite and Jacobi behaviours: expect geometric decay from e^{-x}, corrections from the endpoint. Also the Mellin *preimage* of stratum 2.3 — compute both sides and watch the transform carry structure across.

2.6 Compact interval: Jacobi *control*

$W = (1-x)^\alpha (1+x)^\beta$ on $[-1, 1]$, Jacobi $P_n^{(\alpha,\beta)}$. No ξ connection; included as the pure-endpoint-singularity control. Chebyshev ($\alpha = \beta = \mp\frac{1}{2}$) is the degenerate corner where $W^{3/2}$ is again a Chebyshev weight — the matching matrix should be computable in closed form by hand, a useful sanity anchor.

2.7 Discrete kernels: Charlier, Meixner, Krawtchouk *control / combinatorial*

$W(x) = e^{-a} a^x / x!$ on $\mathbb{N}$ (Charlier $C_n(x; a)$); $\beta^{(x)} c^x / x!$ (Meixner); $\binom{N}{x} p^x q^{N-x}$ (Krawtchouk). Matching matrices are *sums*; integrality is automatic, so the question inverts: which phenomena

(checkerboard, phase law, two-generator structure) survive, and which were artifacts of the contour? Motivated from inside: the MP dual GF factors through a *Charlier kernel* $e^{\sigma+\tau-\sigma\tau}$ (meixner-pollaczek-simple.tex §6) — the discrete family is already embedded in the continuous answer. The conjectured 2-state automaton ($K_0 \leftrightarrow K_1$) should be *provable* here first.

2.8 Genuinely dual: no orthogonal counterpart *new territory*

Once the pairing is bilinear, the kernel need not be symmetric, positive, or even real. Three sub-cases, in increasing wildness:

7a. Unbalanced Γ split. $W = \Gamma(a+ix)\,\Gamma(b-ix)$ with $a \neq b$. Not of the form $|\cdot|^2$: no positive weight, no orthogonal functions — but the moments exist (the 3-Γ lemma *is* the moment generating identity, and it never needed $a = b$), the Hankel determinants can be checked, and the whole dual machinery runs verbatim. The natural first instance: $a = \frac{1}{4}$, $b = \frac{3}{4}$ — pairing the $\Gamma(s/2)$ and $\Gamma(\frac{s+1}{2})$ worlds inside one bilinear form, the dual-basis shadow of the joint sech matrix of 2b.

7b. Classical biorthogonal systems. Systems born biorthogonal: Konhauser pairs (polynomials in x against polynomials in x^k, Laguerre weight); Cauchy biorthogonal polynomials (kernel $\frac{1}{x+y}$, two measures); Laurent biorthogonal polynomials (Toda / continued fractions); Wilson's biorthogonal *rational* functions (${}_9F_8$ level, above the Askey scheme). These test whether our phenomena need the polynomial–times–split form at all, or survive in genuinely two-variable pairings.

7c. ζ-adapted kernels. The endgame. Nothing now forbids kernels built from the object of interest itself: the zeta-basis pairing $\langle F_1, F_2\rangle_\zeta = \int F_1\overline{F_2}\,|\zeta|^{-2}$ already lives here, and one can attempt splits $W = \xi(\frac{1}{2}+ix)\cdot(\cdots)$ where positivity fails on every zero but Hankel nonvanishing may hold. Dual bases are the only framework in which such kernels are legal.

2.9 q-kernels *horizon*

Askey–Wilson $|(e^{2i\theta};q)_\infty / \prod_j (a_j e^{i\theta};q)_\infty|^2$. Natural boundary at $|q| = 1$ — a genuinely new analytic class (kernel not continuable at all). Park here until the classical column is done; coordinate with Rick's Hopf-algebra work if it comes alive.

3 Summary Table

kernel W	polynomials	analytic class	status
e^{-x^2}	Hermite	entire	done
$\lvert\Gamma(\frac{1}{4}+ix)\rvert^2$	MP $\lambda = \frac{1}{4}$	poles, finite dist.	done
$\lvert\Gamma(\frac{3}{4}+ix)\rvert^2$	MP $\lambda = \frac{3}{4}$	poles, finite dist.	**1**
$\operatorname{sech} 2\pi x$ (joint)	MP $\frac{1}{4} \times \frac{3}{4}$	elementary, poles	**1**
$\lvert\Gamma(a+ix)\Gamma(b+ix)\rvert^2$	continuous Hahn	Barnes, 2 pairs	**2**
$e^{-a}a^x/x!$ on $\mathbb{N}$	Charlier	discrete	**3**
$x^\alpha e^{-x}$ on $(0,\infty)$	Laguerre	entire + endpoint	4
$\Gamma(\frac{1}{4}+ix)\Gamma(\frac{3}{4}-ix)$	dual only (7a)	unbalanced, signed	**2**
$\lvert\prod_j \Gamma(a_j+ix)/\Gamma(2ix)\rvert^2$	Wilson	Barnes, 4 pairs	5
$(1-x)^\alpha(1+x)^\beta$	Jacobi	endpoints only	6
Konhauser / Cauchy / rational	born biorthogonal	two-variable pairing	7
ξ-built kernels	dual only (7c)	zeros on the contour	endgame
Askey–Wilson	q-world	natural boundary	horizon

Order of work. (1) $\lambda = \frac{3}{4}$ and the joint sech matrix — verbatim port of existing code, elementary integrals, new arithmetic likely. (2) The unbalanced split $\Gamma(\frac{1}{4}+ix)\Gamma(\frac{3}{4}-ix)$ — first kernel with *no* orthogonal theory; the 3-Γ lemma already covers its moments. (3) Continuous Hahn — does Barnes restore closed forms? (4) Charlier — prove the 2-state combinatorial model where sums replace contours. Then Laguerre/Wilson, Jacobi–Chebyshev as the by-hand anchor, the born-biorthogonal systems, and finally ξ-built kernels.